1

18.（本题满分 12 分）

求不定积分 $\int \frac{dx}{(\sqrt{x}+1)(x+3)}$.

2

(18)(本题满分 12 分)　计算 $\int \frac{x\ln(x+\sqrt{1+x^2})}{(1-x^2)^2}dx$.

3

14. 设连续函数 $f(x)$ 满足 $f(x)+f\left(\frac{1}{x}\right)=2$，则 $\int_{\frac{\pi}{6}}^{\frac{\pi}{3}} \frac{f(\tan^2 x)}{\sin 2x}dx=$ _____.

4

14. $\int_0^1 dx \int_0^{\sqrt{x}} e^{-\frac{y^2}{2}} dy =$ _____.

5

(18)(本题满分 12 分)　设 $f(x)$，$g(x)$ 在 $[-a,a]$ 上连续，若

$f(x)=f(-x)$，$g(x)+g(-x)=m$（m 常数），

（Ⅰ）证明 $\int_{-a}^{a} f(x)g(x)dx=\frac{m}{2}\int_{-a}^{a} f(x)dx$；（Ⅱ）计算 $I=\int_{-\pi}^{\pi} \frac{x\sin x}{(e^x+1)(1+\cos^2 x)}dx$.

6

(18)(本题满分 12 分)　计算定积分 $\int_0^1 \arctan x \cdot \ln(1+x^2)dx$.

7

(20)(本题满分 12 分) 设 $D=\{(x,y)\mid 0\leqslant x\leqslant\sqrt{\pi},0\leqslant y\leqslant\sqrt{\pi}\}$,记

$$I_1=\iint_D\sin(\max\{x^2,y^2\})\mathrm{d}x\mathrm{d}y,\quad I_2=\iint_D\sin(\min\{x^2,y^2\})\mathrm{d}x\mathrm{d}y,$$

8

(Ⅱ) 证明 $\lim\limits_{x\to+\infty}\dfrac{\int_0^x f(t)\mathrm{d}t}{x}=\dfrac{1}{T}\int_0^T f(t)\mathrm{d}t$.

(Ⅲ) 求 $\lim\limits_{x\to+\infty}\dfrac{\int_0^x\frac{\sin^2 t}{1+\cos^2 t}\mathrm{d}t}{\int_0^x\frac{1}{1+\cos^2 t}\mathrm{d}t}$.

9

12. 定积分 $\int_0^1\dfrac{x^2+6x+3}{(x+3)^2+(x^2+x)^2}\mathrm{d}x=$ _____.

10

(12) 已知动点 $M(x,y)$ 在 xOy 平面上运动方程为 $\begin{cases}x=\mathrm{e}^{2t}-t,\\ y=2\mathrm{e}^t\end{cases}$ (时间单位:秒;距离单位:米),则该动点在时间段 $\left[\frac{1}{4}\ln 2,\frac{1}{4}\ln 6\right]$ 内的运动路程是_____.

18.【解】 令$\sqrt{x}=t$,可知$x=t^2$,且$\mathrm{d}x=2t\mathrm{d}t$,于是

1

$$\int\frac{\mathrm{d}x}{(\sqrt{x}+1)(x+3)}=\int\frac{2t}{(t+1)(t^2+3)}\mathrm{d}t.$$

设

$$\frac{2t}{(t+1)(t^2+3)}=\frac{A}{t+1}+\frac{Bt+C}{t^2+3}=\frac{(A+B)t^2+(B+C)t+(3A+C)}{(t+1)(t^2+3)},$$

即

$$A+B=0,B+C=2,3A+C=0,$$

解得$A=-\frac{1}{2},B=\frac{1}{2},C=\frac{3}{2}$,从而

$$\begin{aligned}\int\frac{2t}{(t+1)(t^2+3)}\mathrm{d}t&=-\frac{1}{2}\int\frac{1}{1+t}\mathrm{d}t+\frac{1}{2}\int\frac{t+3}{t^2+3}\mathrm{d}t\\&=-\frac{1}{2}\ln(1+t)+\frac{1}{2}\int\frac{t}{t^2+3}\mathrm{d}t+\frac{3}{2}\int\frac{1}{t^2+3}\mathrm{d}t\\&=-\frac{1}{2}\ln(1+t)+\frac{1}{4}\ln(t^2+3)+\frac{\sqrt{3}}{2}\arctan\frac{t}{\sqrt{3}}+C\\&=-\frac{1}{2}\ln(1+\sqrt{x})+\frac{1}{4}\ln(x+3)+\frac{\sqrt{3}}{2}\arctan\sqrt{\frac{x}{3}}+C.\end{aligned}$$

2 解 原式$=\frac{1}{2}\int\ln(x+\sqrt{1+x^2})\mathrm{d}\left(\frac{1}{1-x^2}\right)=\frac{1}{2}\frac{\ln(x+\sqrt{1+x^2})}{1-x^2}+\frac{1}{2}\int\frac{1}{(x^2-1)\sqrt{1+x^2}}\mathrm{d}x$,

而$\int\frac{1}{(x^2-1)\sqrt{1+x^2}}\mathrm{d}x\overset{x=\tan t}{=}\int\frac{\cos t}{\sin^2t-\cos^2t}\mathrm{d}t=\frac{1}{2\sqrt{2}}\ln\left|\frac{\sqrt{2}\sin t-1}{\sqrt{2}\sin t+1}\right|+C$

$=\frac{1}{2\sqrt{2}}\ln\left|\frac{\sqrt{2}x-\sqrt{1+x^2}}{\sqrt{2}x+\sqrt{1+x^2}}\right|+C$,因而

原式$=\frac{1}{2}\frac{\ln(x+\sqrt{1+x^2})}{1-x^2}+\frac{1}{4\sqrt{2}}\ln\left|\frac{\sqrt{2}x-\sqrt{1+x^2}}{\sqrt{2}x+\sqrt{1+x^2}}\right|+C.$

3

$$\begin{aligned}&\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\frac{f\left(\tan^2x\right)}{\sin2x}\,\mathrm{d}x\overset{t=\frac{\pi}{2}-x}{=\!=\!=}\int_{\frac{\pi}{3}}^{\frac{\pi}{6}}\frac{f\left[\tan^2\left(\frac{\pi}{2}-t\right)\right]}{\sin2\left(\frac{\pi}{2}-t\right)}\mathrm{d}\left(\frac{\pi}{2}-t\right)\\&=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\frac{f\left(\cot^2x\right)}{\sin2x}\,\mathrm{d}x=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\frac{f\left(\frac{1}{\tan^2x}\right)}{\sin2x}\mathrm{d}x\\&=\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\left[\frac{f\left(\tan^2x\right)+f\left(\frac{1}{\tan^2x}\right)}{\sin2x}\right]\mathrm{d}x\\&=\frac{1}{2}\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\frac{2}{\sin2x}\,\mathrm{d}x=\int_{\frac{\pi}{6}}^{\frac{\pi}{3}}\csc2x\,\mathrm{d}x\\&=-\frac{1}{2}\ln|\csc2x+\cot2x|\Big|_{\frac{\pi}{6}}^{\frac{\pi}{3}}\\&=-\frac{1}{2}\left[\ln\left(\frac{2}{\sqrt{3}}-\frac{1}{\sqrt{3}}\right)-\ln\left(\frac{2}{\sqrt{3}}+\frac{1}{\sqrt{3}}\right)\right]\\&=\frac{\ln3}{2}.\end{aligned}$$

14.【答案】 $\mathrm{e}^{-\frac{1}{2}}$ 4

【分析】 交换积分次序得

$$\begin{aligned}I&=\int_0^1\mathrm{d}y\int_{y^2}^1\mathrm{e}^{-\frac{y^2}{2}}\mathrm{d}x=\int_0^1(1-y^2)\mathrm{e}^{-\frac{y^2}{2}}\mathrm{d}y=\int_0^1\mathrm{e}^{-\frac{y^2}{2}}\mathrm{d}y+\int_0^1y\mathrm{d}\left(\mathrm{e}^{-\frac{y^2}{2}}\right)\\&=\int_0^1\mathrm{e}^{-\frac{y^2}{2}}\mathrm{d}y+y\mathrm{e}^{-\frac{y^2}{2}}\Big|_0^1-\int_0^1\mathrm{e}^{-\frac{y^2}{2}}\mathrm{d}y=\mathrm{e}^{-\frac{1}{2}}.\end{aligned}$$

5 (Ⅱ)取$f(x)=\frac{x\sin x}{\cos^2x+1},g(x)=\frac{1}{\mathrm{e}^x+1}$,则$f(-x)=f(x),g(x)+g(-x)=1$.

由(Ⅰ)知

$$I=\frac{1}{2}\int_{-\pi}^{\pi}\frac{x\sin x}{\cos^2x+1}\mathrm{d}x=\int_0^{\pi}\frac{x\sin x}{\cos^2x+1}\mathrm{d}x=\frac{\pi}{2}\int_0^{\pi}\frac{\sin x}{\cos^2x+1}\mathrm{d}x=-\frac{\pi}{2}\int_0^{\pi}\frac{\mathrm{d}\cos x}{\cos^2x+1}$$

$$=-\frac{\pi}{2}\arctan\cos x\Big|_0^{\pi}=-\frac{\pi}{2}\left[-\frac{\pi}{4}-\frac{\pi}{4}\right]=\frac{\pi^2}{4}.$$

解一 $\int\ln(1+x^2)\mathrm{d}x=x\ln(1+x^2)-\int\frac{2x^2}{1+x^2}\mathrm{d}x$

$=x\ln(1+x^2)-2\int\frac{x^2+1-1}{1+x^2}\mathrm{d}x=x\ln(1+x^2)-2x+2\arctan x+C_1$

故原式$=\int_0^1\arctan x\mathrm{d}[x\ln(1+x^2)-2x+2\arctan x]$

$=\int_0^1\arctan x\mathrm{d}[x\ln(1+x^2)-2x]+2\int_0^1\arctan x\mathrm{d}(\arctan x)$

$=\arctan x\cdot[x\ln(1+x^2)-2x]\,|_0^1-\int_0^1[x\ln(1+x^2)-2x]\cdot\frac{1}{1+x^2}\mathrm{d}x+(\arctan^2x)\,|_0^1$

$=\frac{\pi}{4}(\ln2-2)+\frac{\pi^2}{16}-\int_0^1\frac{x}{1+x^2}\ln(1+x^2)\mathrm{d}x+\int_0^1\frac{2x}{1+x^2}\mathrm{d}x$

6

$=\frac{\pi}{4}(\ln2-2)+\frac{\pi^2}{16}-\frac{1}{4}[\ln(1+x^2)]^2\,|_0^1+\ln(1+x^2)\,|_0^1$

$=\frac{\pi}{4}(\ln2-2)+\frac{\pi^2}{16}-\frac{1}{4}\ln^22+\ln2.$

解二 $\int\arctan x\mathrm{d}x=x\arctan x-\int\frac{x}{1+x^2}\mathrm{d}x=x\arctan x-\frac{1}{2}\ln(1+x^2)+C.$

故原式$=\int_0^1\arctan x\cdot\ln(1+x^2)\mathrm{d}x=\int_0^1\ln(1+x^2)\mathrm{d}\left[x\arctan x-\frac{1}{2}\ln(1+x^2)\right]$

$=\int_0^1\ln(1+x^2)\mathrm{d}(x\arctan x)-\frac{1}{2}\int_0^1\ln(1+x^2)\mathrm{d}(\ln(1+x^2))$

$=x\arctan x\cdot\ln(1+x^2)\,|_0^1-2\int_0^1\frac{x^2}{1+x^2}\arctan x\mathrm{d}x-\frac{1}{4}\ln^2(1+x^2)\,|_0^1$

$=\frac{\pi}{4}\ln2-2\int_0^1\left(\arctan x-\frac{\arctan x}{1+x^2}\right)\mathrm{d}x-\frac{1}{4}\ln^22$

$=\frac{\pi}{4}\ln2-2\int_0^1\arctan x\mathrm{d}x+\arctan^2x\,|_0^1-\frac{1}{4}\ln^22$

$=\frac{\pi}{4}\ln2-2\left[x\arctan x-\frac{1}{2}\ln(1+x^2)\right]|_0^1+\frac{\pi^2}{16}-\frac{1}{4}\ln^22$

$=\frac{\pi}{4}\ln2-\frac{\pi}{2}+\ln2+\frac{\pi^2}{16}-\frac{1}{4}\ln^22.$

7

证明 $I_2 < I_1$.

证　记 $D_0 = \{(x,y) \mid 0 \leqslant y \leqslant x \leqslant \sqrt{\pi}\}$. 根据轮换对称性,有

$$I_1 = 2\iint_{D_0} \sin(\max\{x^2,y^2\})\,dx\,dy = 2\iint_{D_0} \sin x^2\,dx\,dy$$

$$= 2\int_0^{\sqrt{\pi}} dx \int_0^x \sin x^2\,dy = 2\int_0^{\sqrt{\pi}} x\sin x^2\,dx = -\cos x^2 \Big|_0^{\sqrt{\pi}} = 2.$$

$$I_2 = 2\iint_{D_0} \sin(\min\{x^2,y^2\})\,dx\,dy = 2\iint_{D_0} \sin y^2\,dx\,dy$$

$$= 2\int_0^{\sqrt{\pi}} dy \int_y^{\sqrt{\pi}} \sin y^2\,dx = 2\int_0^{\sqrt{\pi}} (\sqrt{\pi}-y)\sin y^2\,dy = 2\sqrt{\pi}\int_0^{\sqrt{\pi}} \sin y^2\,dy - 2.$$

又因为 $\int_0^{\sqrt{\pi}} \sin y^2\,dy \overset{t=y^2}{=} \int_0^{\pi} \sin t\,d(\sqrt{t}) = \sqrt{t}\sin t \Big|_0^{\pi} - \int_0^{\pi} \sqrt{t}\cos t\,dt$

$$= -\int_0^{\frac{\pi}{2}} \sqrt{t}\cos t\,dt - \int_{\frac{\pi}{2}}^{\pi} \sqrt{t}\cos t\,dt = -\int_0^{\frac{\pi}{2}} \sqrt{t}\cos t\,dt - \int_0^{\frac{\pi}{2}} \sqrt{\pi-t}\cos(\pi-t)\,dt$$

$$= \int_0^{\frac{\pi}{2}} \cos t \cdot (\sqrt{\pi-t} - \sqrt{t})\,dt \overset{t=\frac{\pi}{2}-x}{=} \int_0^{\frac{\pi}{2}} \sin x \cdot \left(\sqrt{\frac{\pi}{2}+x} - \sqrt{\frac{\pi}{2}-x}\right)dx$$

$$= \int_0^{\frac{\pi}{2}} \sin x \cdot \frac{2x}{\sqrt{\frac{\pi}{2}+x}+\sqrt{\frac{\pi}{2}-x}}\,dx < \int_0^{\frac{\pi}{2}} \sin x \cdot \frac{2x}{\sqrt{\pi}}\,dx = \frac{2}{\sqrt{\pi}}.$$

所以 $I_2 < 2\sqrt{\pi}\cdot\frac{2}{\sqrt{\pi}} - 2 = 2 = I_1$.

8

(Ⅲ) $\int_0^{\pi} \frac{1}{1+\cos^2 t}\,dt = \int_{-\frac{\pi}{2}}^{\frac{\pi}{2}} \frac{1}{1+\cos^2 t}\,dt = 2\int_0^{\frac{\pi}{2}} \frac{1}{1+\cos^2 t}\,dt = 2\int_0^{\frac{\pi}{2}} \frac{1}{\sec^2 t+1}\,d(\tan t)$

$$= 2\int_0^{\frac{\pi}{2}} \frac{1}{\tan^2 t+2}\,d(\tan t) = \frac{2}{\sqrt{2}}\arctan\frac{\tan t}{\sqrt{2}}\Big|_0^{\frac{\pi}{2}} = \frac{\sqrt{2}}{2}\pi,$$

$$\int_0^{\pi} \frac{\sin^2 t}{1+\cos^2 t}\,dt = \int_0^{\pi} \frac{1-\cos^2 t}{1+\cos^2 t}\,dt = 2\int_0^{\pi} \frac{1}{1+\cos^2 t}\,dt - \pi = (\sqrt{2}-1)\pi,$$

$$\lim_{x\to+\infty} \frac{\int_0^x \frac{\sin^2 t}{1+\cos^2 t}\,dt}{\int_0^x \frac{1}{1+\cos^2 t}\,dt} = \lim_{x\to+\infty} \frac{\frac{1}{x}\int_0^x \frac{\sin^2 t}{1+\cos^2 t}\,dt}{\frac{1}{x}\int_0^x \frac{1}{1+\cos^2 t}\,dt} = \frac{\frac{1}{\pi}\int_0^{\pi} \frac{\sin^2 t}{1+\cos^2 t}\,dt}{\frac{1}{\pi}\int_0^{\pi} \frac{1}{1+\cos^2 t}\,dt} = \frac{\sqrt{2}-1}{\frac{\sqrt{2}}{2}} = 2-\sqrt{2}.$$

9

12.【答案】 $\arctan\frac{1}{2}$

【分析】 分子分母同除以 $(x+3)^2$,得

$$\int_0^1 \frac{x^2+6x+3}{(x+3)^2+(x^2+x)^2}\,dx = \int_0^1 \frac{\frac{x^2+6x+3}{(x+3)^2}}{1+\left(\frac{x^2+x}{x+3}\right)^2}\,dx$$

$$= \int_0^1 \frac{d\left(\frac{x^2+x}{x+3}\right)}{1+\left(\frac{x^2+x}{x+3}\right)^2} = \arctan\left(\frac{x^2+x}{x+3}\right)\Big|_0^1$$

$$= \arctan\frac{1}{2}.$$

10

解　$s = \int_{\frac{1}{4}\ln 2}^{\frac{1}{4}\ln 6} \sqrt{x'^2(t)+y'^2(t)}\,dt = \int_{\frac{1}{4}\ln 2}^{\frac{1}{4}\ln 6} \sqrt{(2e^{2t}-1)^2+4e^{2t}}\,dt = \int_{\frac{1}{4}\ln 2}^{\frac{1}{4}\ln 6} \sqrt{1+4e^{4t}}\,dt$, 令 $\sqrt{1+4e^{4t}}$ u, 则

$$s = \frac{1}{2}\int_3^5 \frac{u^2}{u^2-1}\,du = 1+\frac{1}{4}\ln\frac{4}{3}.$$